

Type III Collagen Glomerulopathy (Collagenofibrotic Glomerulopathy)

Agnes Fogo, MD

The AJKD Atlas of Renal Pathology presents a compilation of figures on a specific pathologic entity. You may download the figures to create your own personal, non-commercial library of images or to create slides for teaching purposes.

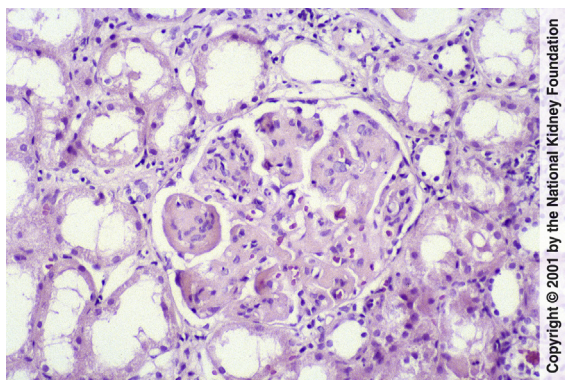


Fig 1. Collagenofibrotic glomerulopathy, also known as type III collagen glomerulopathy, is an apparent autosomal recessive inherited condition with abnormal accumulation of type III collagen in glomeruli, often associated with hemolytic uremic syndrome. In rare cases, systemic type III collagen deposits have been noted. The glomeruli appear enlarged with a lobular or nodular configuration by light microscopy with pale amorphous expansion of the mesangial matrix. (Hematoxylin & eosin, $\times 200$).

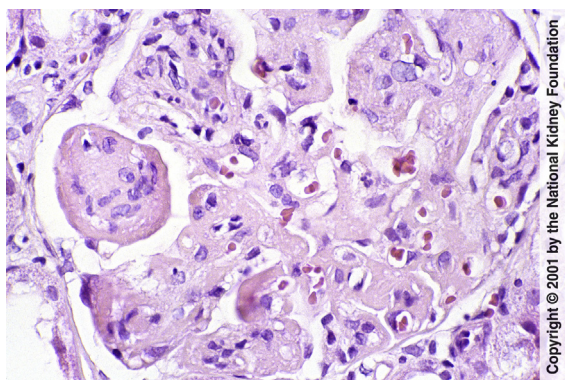


Fig 2. This slide shows collagenofibrotic glomerulopathy (also known as type III collagen glomerulopathy) on higher power, illustrating the lobular to nodular appearance, with amorphous pale slightly eosinophilic material expanding mesangial areas and filling capillary loops. There is also moderate mesangial hypercellularity. (Hematoxylin & eosin, $\times 400$).

From the Department of Pathology, Vanderbilt University Medical Center, Nashville, TN.

Medical Photographer: Brent Weedman.

Address author queries to Agnes Fogo, MD, Department of Pathology, Vanderbilt University Medical Center, MCN C-3310, Nashville, TN 37232. E-mail: Agnes.Fogo@vanderbilt.edu

Am J Kidney Dis. 38(4):E15-E16.

© 2001 by the National Kidney Foundation, Inc. Published by Elsevier Inc. All rights reserved.

0272-6386/\$36.00

[http://dx.doi.org/10.1053/S0272-6386\(13\)90045-7](http://dx.doi.org/10.1053/S0272-6386(13)90045-7)

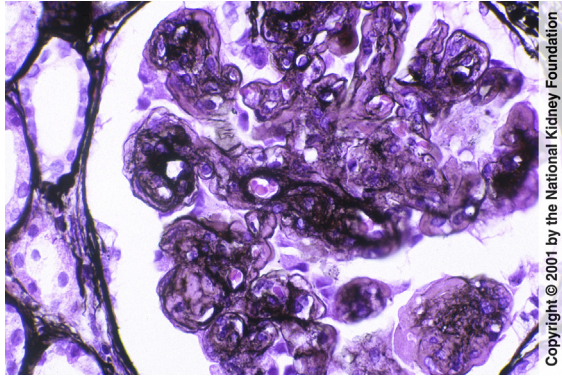


Fig 3. The lobular to nodular appearance of collagenofibrotic glomerulopathy (also known as type III collagen glomerulopathy) is shown here, with a membranoproliferative pattern suggested by the splitting of the glomerular basement membrane. However, standard immunofluorescent studies reveal the absence of immune complexes (not shown). Research studies for type III collagen have demonstrated the nature of the deposits, which may also be seen by electron microscopy (see Figure 9). (Jones' silver stain, $\times 400$).

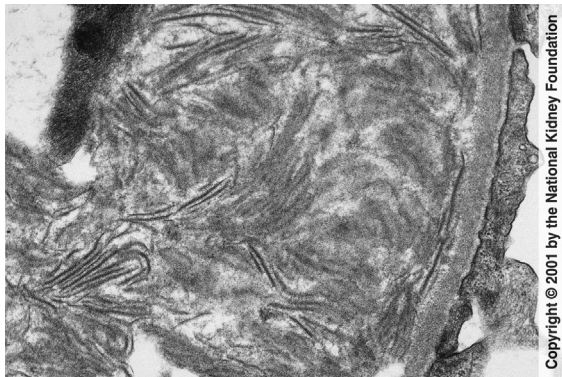


Fig 4. The classic deposits of collagenofibrotic glomerulopathy (also known as type III collagen glomerulopathy) are shown here. The deposits have a whorling appearance on low power and are abundant with cross-striated appearance on high power, typical of the fibrillary type III collagen. The rare fibrillary collagen that may be part of sclerosing processes in a variety of diseases should not be mistaken for collagenofibrotic glomerulopathy, where there is a large abundance of this type of collagen fibrils in the mesangium and also in the subendothelium. (Transmission electron microscopy, $\times 10000$).